Md. Zubayer Ahmad Shibly

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Portfolio

About Me

"Know Thyself" guides my journey as a Computer Science and Engineering student focused on Artificial Intelligence and Machine Learning, with interests in Deep Learning, Transformers, Explainable AI, data-driven modeling, and AI agents. I value self-discovery and continuous growth, and aspire to become an AI researcher solving real-world problems through practical, impactful research. My experience includes applied AI in healthcare, network security, and decision-support systems, with hands-on work across the machine learning lifecycle and building agentic intelligent systems.

Education

Southeast University (SEU), Tejgaon, Dhaka, Bangladesh July 2022 – Present
B.Sc. in Computer Science and Engineering

- **CGPA:** 4.00/4.00
- **Thesis:** FedGCF-Net: Federated Genetically-Optimized Contrast-Enhanced Fusion Network for Brain Tumor Classification. **Manuscript under review at International Journal of Intelligent Systems, WILEY**
- **Coursework:** Artificial Intelligence; Introduction to Data Mining; Cryptography & Network Security; Data Structures; Algorithms; Theory of Computing; Database Design; Linear Algebra; Probability & Statistics

Research Experience

Collaborative Research Projects 2026 – Present
Supervisor: Dr. Maybin Muyeba, University of Salford, United Kingdom
School of Science, Engineering & Environment

- Developing an explainable federated intrusion detection framework based on Empirical Bernstein certified Bandit-KL mixing with tree stacking. **Accepted at IEEE FUZZ 2026, Netherlands - B-core conference**
- Designing FedBridge++, a privacy-preserving federated self-supervision framework for tabular loan approval using interpretable transformers. **Accepted at SERA 2026, USA - C-core conference**

Collaborative Research Projects 2025 – April 2026
Supervisor: Rajon Bardhan, Augusta University, USA
School of Computer and Cyber Sciences

- UADGE-FS: Uncertainty-Aware, Dominance-Gated Evolutionary Feature Selection. **Target Conference: ICDM 2026, China - A* conference**
- TRUST-GFS: Tie-safe Reliability and Uplift Screening for Target-guided Genetic Feature Selection, a robust target-guided feature selection framework. **Submitted to NeurIPS 2026 - A* conference**

Publications and Research Manuscripts

Accepted Conference Papers (Forthcoming)

- Leveraging Multi-Algorithmic Feature Selection and Ensemble Machine Learning for Accurate Detection of Polycystic Ovarian Syndrome. **IEEE ICCCNT 2025, Indore, India**
- A Novel Hybrid Feature Selection and Ensemble Learning Approach for Mortality Risk Classification in Hepatitis B Patients: An Explainable AI Study. **ICCIT 2025, Cox's Bazar, Bangladesh**
- Explainable AI-Driven Ensemble Learning Framework for PCOS Diagnosis Using AIM-PDCF and Quantum-GraphRFE Feature Selection. **ICCIT 2025, Cox's Bazar, Bangladesh**

Conference Papers Under Review

- Adaptive Biomarker Categorization with DiCAT and TCCA Feature Selection Model for Robust, Explainable Early Ovarian Cancer Detection. **IRAI 2026, Melbourne, Australia**

- An Explainable Stacking Ensemble Framework with Disagreement-Aware Feature Selection for Loan Approval Prediction. **TENSYMP2026, Penang, Malaysia**
- An Interpretable Framework for Early Lung Cancer Prediction Using Graph Fusion and Role Stability for Robust Feature Selection with TabFGT. **AVSS2026, Lecce, Italy - B-core conference**

Journal Articles Under Review

- Q-SEAL: An Explainable Transformer-Based Meta-Ensemble Model for Ovarian Cancer Prediction and Deployment Evaluation. **Journal of Pathology Informatics**
- Explainable Ovarian Cancer Diagnosis: Fusion-Based RFE-SHAP and Quantum-Inspired ANN-QUBO Feature Selection Model with Two-Tier Evaluation. **Digital Health (SAGE)**
- AIDPCP: An Adaptive Intelligent Preprocessing and Clustering Pipeline for Obesity Prediction with Explainable AI. **IET Software (Wiley)**

Professional Experience

NeuroCrack Technologies — AI Agent Developer Intern (Offer Letter) March 2026 – Present

Roots of Rise — Director of Project Management (Volunteer - NGO) Sep 2025 – Present

IMUN — Campus Ambassador Intern (Certificate | Offer Letter | Conference) Mar 2025 – May 2025

Laga Tour — Project Manager (Tech) (Part-time) Feb 2024 – Apr 2024

Projects

MultiDx Clinical AI: Multi-Disease Prediction System Southeast University, 2025

Tools used: Python, FastAPI, PyTorch, scikit-learn, Next.js, TypeScript | Live Demo | GitHub

- Developed a full-stack clinical AI platform with an end-to-end workflow from modeling to deployment.
- Built and integrated four disease-specific models for Ovarian Cancer, PCOS, Hepatitis B, and Brain Tumor (MRI) prediction.
- Incorporated explainability modules using SHAP and LIME with PDP, ICE, and ALE based analytical views.

AniMaze: AI-Powered Anime & Manga Platform Team Project, 2025

Tools used: TypeScript, MySQL, Supabase | GitHub

- Built a full-stack platform with a responsive interface, structured navigation, and user-focused page design.
- Added AI-powered quiz generation, XP-based progression, leaderboard functionality, and typo-tolerant smart search.
- Implemented role-based dashboards, external API integrations, and subscription badge logic for access control.

Arduino Home Security and Fire/Smoke Detection System with Automatic Water Pump Activation Southeast University, 2026

Tools used: Arduino Uno, HC-05 Bluetooth, MQ-2 & Flame Sensor, Servo, Relay, Keypad | GitHub

- Developed a keypad and LCD based door security system with servo-controlled locking and timed auto re-lock.
- Enabled Bluetooth-based mobile control for gate access, pump operation, and alarm silencing.
- Implemented fire and smoke detection that automatically triggers alarms and activates the water pump via relay.

Achievements

- **Department Rank:** Ranked 1st in the Department of CSE, Southeast University, based on academic performance.
- **Full Tuition Waiver:** Awarded a full scholarship covering all four years of the B.Sc. program in CSE at Southeast University.
- **Harvard HSIL Hackathon 2026:** Top 22 of 650 applicant teams as part of Team Precision Health Innovators, Dhaka Hub, April 10–11 | Certificate
- **Leadership Recognition:** Aspire Institute Awards by Aspire Leaders Program 2025 | Certificate

Workshops

- Webinar on Programming Contest & Research — IEEE Computer Society, SEU Student Branch Chapter (Dec 2025) | Certificate
- LaTeX Unlocked: The Research Writing Workshop — IEEE Computer Society, SEU Student Branch Chapter (Dec 2025) | Certificate

Certifications

- Advanced Research in Machine Learning & XAI — Neural Research (Certificate) Jun 2025
- Python Functions, Files, and Dictionaries — University of Michigan (Coursera) (Certificate) Mar 2025
- Foundations of Project Management — Coursera (Certificate) Jun 2024
- Cybersecurity Foundations — LinkedIn Learning (Certificate) Feb 2024
- ENGL210: Technical Writing — Saylor Academy (Certificate) Feb 2024

Technologies / Skills

AI/ML: Deep Learning, Federated Learning, Transformers, AI Agent, Explainable AI, Feature Engineering

Data: Data Analysis, Dataset Collection, Data Preprocessing, Statistics

Databases: MySQL, MongoDB

Tools: PyTorch, scikit-learn, NumPy, Pandas, Matplotlib, Jupyter Notebook, Google Colab, Git, Overleaf

Programming: Python, C, Java, Dart

Productivity: Excel/Google Sheets, Word/Google Docs, PowerPoint

Extracurricular Activities

BYLCx — Volunteerism for Social Change Mar 2024

- Completed training on volunteering fundamentals, social responsibility, and community impact | Certificate

FICC (Farees International Career Counseling) — Debate Competition Winner Feb 2020

- Led the team to victory as team president | Certificate

References

- **Shifat Ahmed**. Coordinator & Assistant Professor, Dept. of CSE, Southeast University
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  Phone: +880 1717-691945
- **Khandaker Mohammad Mohi Uddin**. Assistant Professor, Dept. of CSE, Southeast University
  Email: mohiuddin.kh@seu.edu.bd
  Phone: +880 1520-102813
  Undergraduate Thesis Supervisor